

Solution Code - JUnit, Mockito and SL4J - JUnit_Spring_Test_MockMvc

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\test\java\com\example\AppTest.java

```
1 | package com.example;
2 |
3 | import org.junit.jupiter.api.Test;
4 | import org.springframework.beans.factory.annotation.Autowired;
5 | import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
6 | import org.springframework.boot.test.mock.mockito.MockBean;
7 | import org.springframework.test.web.servlet.MockMvc;
8 | import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
9 | import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;
10 |
11 | @WebMvcTest
12 | public class UserControllerTest {
13 |     @Autowired
14 |     private MockMvc mockMvc;
15 |
16 |     @MockBean
17 |     private UserService userService;
18 |
19 |     @Test
20 |     public void testGetUser() throws Exception {
21 |         mockMvc.perform(get("/users/1")).andExpect(status().isOk());
22 |     }
23 | }
```

Source File: src\test\java\com\example\UserControllerTest.java

```
1 | package com.example;
2 |
3 | import org.junit.jupiter.api.Test;
4 | import org.springframework.beans.factory.annotation.Autowired;
5 | import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
6 | import org.springframework.test.web.servlet.MockMvc;
7 | import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
8 | import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;
9 |
10 | @WebMvcTest
11 | public class UserControllerTest {
12 |     @Autowired
13 |     private MockMvc mockMvc;
14 |
15 |     @Test
16 |     public void testGetStatus() throws Exception {
17 |         mockMvc.perform(get("/api/status"))
```

Solution Code - JUnit, Mockito and SL4J - JUnit_Spring_Test_MockMvc

```
18 |         .andExpect(status().isOk())
19 |         .andExpect(content().string("API is running"));
20 |     }
21 | }
```